

Practice 1: Operation clues

Name: _____ Date: _____

Evaluate each expression. Show the order in which you calculate.

1. $8 + 3 \times 4$

Work space

2. $24 - 12 \div 3$

Work space

3. $7 \times 5 + 6$

Work space

Practice 2: Parentheses first

Name: _____ Date: _____

Evaluate each expression. Calculate inside parentheses first.

1. $(8 + 3) \times 4$

Work space

2. $24 - (12 \div 3)$

Work space

3. $7 \times (5 + 6)$

Work space

Practice 3: Multiplication and division together

Name: _____ Date: _____

Work from left to right when multiplication and division have equal priority.

1. $36 \div 4 \times 3$

Work space

2. $5 \times 6 \div 2$

Work space

3. $48 \div 6 \times 2 + 1$

Work space

Practice 4: Addition and subtraction together

Name: _____ Date: _____

Work from left to right when addition and subtraction have equal priority.

1. $32 - 7 + 5$

Work space

2. $18 + 9 - 6$

Work space

3. $50 - 14 + 8 - 6$

Work space

Practice 5: Mixed operations

Name: _____ Date: _____

Evaluate. Underline the operation you do first.

1. $9 + 18 \div 3$

Work space

2. $6 \times 7 - 20$

Work space

3. $45 \div 5 + 4 \times 3$

Work space

Practice 6: Multi-step expressions

Name: _____ Date: _____

Show every step and keep the expression equal at each step.

1. $3 \times (12 - 5) + 8$

Work space

2. $64 \div (4 + 4) + 7$

Work space

3. $40 - 2 \times (6 + 3)$

Work space

Practice 7: Write an expression

Name: _____ Date: _____

Write an expression for each phrase, then evaluate it.

1. Add 6 to the product of 4 and 8.

Work space

2. Subtract 9 from the quotient of 72 and 8.

Work space

3. Multiply the sum of 5 and 7 by 3.

Work space

Practice 8: Grouping symbols

Name: _____ Date: _____

Add parentheses to make each statement true, then check it.

1. Add parentheses to $6 + 4 \times 5$ so its value is 50.

Work space

2. Add parentheses to $18 - 6 \div 3$ so its value is 4.

Work space

3. Add parentheses to $2 + 8 \times 3 - 1$ so its value is 29.

Work space

Practice 9: One-step addition equations

Name: _____ Date: _____

Find the unknown. Check by substituting your answer.

1. $x + 27 = 65$

Work space

2. $148 + n = 300$

Work space

3. $m + 425 = 1,000$

Work space

Practice 10: One-step subtraction equations

Name: _____ Date: _____

Find the unknown and check your solution.

1. $y - 19 = 46$

Work space

2. $92 - p = 37$

Work space

3. $q - 275 = 425$

Work space

Practice 11: One-step multiplication equations

Name: _____ Date: _____

Use the inverse operation to find each unknown.

1. $4a = 36$

Work space

2. $7b = 84$

Work space

3. $12c = 144$

Work space

Practice 12: One-step division equations

Name: _____ Date: _____

Use multiplication to check each answer.

1. $d \div 5 = 9$

Work space

2. $e \div 8 = 11$

Work space

3. $156 \div f = 12$

Work space

Practice 13: Two-step equations

Name: _____ Date: _____

Undo the operations in reverse order. Check your answer.

1. $2x + 5 = 19$

Work space

2. $3y - 4 = 20$

Work space

3. $n \div 4 + 6 = 15$

Work space

Practice 14: Equations with parentheses

Name: _____ Date: _____

Solve each equation by simplifying the grouped part first.

1. $(x + 3) \times 4 = 28$

Work space

2. $5(p - 2) = 35$

Work space

3. $q + (6 \times 7) = 80$

Work space

Practice 15: Translate and solve

Name: _____ Date: _____

Write an equation for each situation, then solve it.

1. A number plus 38 is 91. What is the number?

Work space

2. Six equal packs contain 54 pencils. How many pencils per pack?

Work space

3. A number decreased by 24 is 57. Find the number.

Work space

Practice 16: Word problems with expressions

Name: _____ Date: _____

Write an expression before solving each problem.

1. Four buses carry 36 students each, and 18 students arrive by car. How many students are there?

Work space

2. Maya has 60. She spends \$14, then divides the rest equally among 2 savings jars. How much goes in each jar?

Work space

3. A baker makes 5 trays of 24 rolls and gives away 17 rolls. How many remain?

Work space

Practice 17: Compare expressions

Name: _____ Date: _____

Evaluate both expressions. Use $>$, $<$, or $=$.

1. $6 + 4 \times 5$ ____ $(6 + 4) \times 5$

Work space

2. $48 \div 6 + 2$ ____ $48 \div (6 + 2)$

Work space

3. $3 \times (9 - 2)$ ____ $3 \times 9 - 2$

Work space

Practice 18: Properties and equivalent expressions

Name: _____ Date: _____

Use a property or the order of operations to explain each result.

1. Rewrite $7 \times (20 + 3)$ using the distributive property, then evaluate.

Work space

2. Are 4×9 and 9×4 equivalent? Name the property.

Work space

3. Add parentheses to $12 + 3 \times 4$ to make an expression equal to 60.

Work space

Practice 19: Mixed review

Name: _____ Date: _____

Solve carefully. Show enough work for someone else to follow.

1. $90 - 4 \times (8 + 2)$

Work space

2. $3r + 7 = 31$

Work space

3. A club has 7 groups of 12 members. It sends 15 members home early. How many remain?

Work space

Practice 20: Challenge and explain

Name: _____ Date: _____

Solve and explain your order of operations or equation steps.

1. Find the missing number: $5 \times (x + 2) = 45$.

Work space

2. Write two different expressions equal to 32 using at least two operations in each.

Work space

3. Leo says $8 + 12 \div 4 = 5$. Explain his mistake and give the correct value.

Work space